

Clinical Effects of a Controlled Trial of Methylphenidate on Adolescents with Attention Deficit Disorder

RAFAEL KLORMAN, PH.D., JOAN T. BRUMAGHIM, PH.D., PATRICIA A. FITZPATRICK, PH.D.,
AND AGNETA D. BORGSTEDT, M.D.

Abstract. Forty-eight attention deficit disorder patients, 12 to 18 years old and without previous stimulant therapy, received a double-blind trial of methylphenidate and placebo for 3 weeks each. Stimulant treatment produced mild side effects and weight reduction. Methylphenidate significantly reduced teachers' and parents' ratings of hyperactivity, inattention, and oppositionality. In addition, patients rated themselves as clinically improved and reported elevated subjective mood during stimulant therapy. Treatment benefits were comparable for patients with and without concurrent conduct or oppositional disorder as well as those with and without past or present depressive disorders. These results support the continued effectiveness of stimulant therapy for attention deficit disorder in adolescence. However, the magnitude of clinical effectiveness reported was smaller than previously found in younger patients. *J. Am. Acad. Child Adolesc. Psychiatry*, 1990, 29, 5:702-709. **Key Words:** attention deficit disorder, methylphenidate, adolescence.

The effectiveness of stimulant therapy for attention deficit disorder (ADD) in childhood has been extensively documented (Gittelman and Kanner, 1986). In contrast, there is a dearth of research evaluating the utility of these medications for ADD adolescents, a significant proportion of whom display full or residual ADD as well as conduct disorders and academic difficulties (Hechtman et al., 1984; Gittelman et al., 1985; Klorman, 1986). Open clinical trials with stimulants have found clinical improvement and no evidence of medication abuse in ADD adolescents (e.g., Lerer and Lerer, 1977). In a controlled trial of methylphenidate (0.3 and 0.6 mg/Kg) with a sample of stimulant-responsive ADD adolescents, Varley (1983) found significant improvement over placebo in parents' and teachers' ratings. Klorman et al. (1987) found similar benefits of methylphenidate in a double-blind trial of methylphenidate (40 mg/day) involving adolescent ADD patients, most of whom had responded positively to methylphenidate in childhood. In addition, stimulant therapy improved these patients' accuracy and speed in Continuous Performance and Sternberg memory scanning tests and increased the amplitude of the P3b component of event-related potentials obtained during these tasks (Coons et al., 1987).

These results suggest that stimulant therapy has clinical and cognitive benefits for ADD in adolescence, as previously found in childhood. Because previous controlled studies involved primarily stimulant responders, the authors investigated whether unselected, previously untreated ADD adolescents benefit from this therapy.

Method

Subjects

Subjects were 48 patients (42 males, 46 Caucasian) 12 to 18 years old ($\bar{X} = 14.12 \pm 1.69$) referred to this project between May 1984 and April 1988 for evaluation of clinical response to stimulant therapy. Fifteen of the patients were referred by a pediatric neurologist (ADB), and the remainder by pediatricians, pediatric neurologists, child psychiatrists, or school psychologists. Two patients had received brief (2 to 4 months) trials of stimulants in childhood, whereas the remaining 46 patients had none. No patient had ever been treated with any other psychotropics. All subjects fulfilled criteria for *DSM-III* diagnoses of ADD for childhood and for the period preceding referral, as determined in separate standardized psychiatric interviews of both the parent(s) and the patient. A clinical psychologist (RK) administered the entire Diagnostic Instrument for Childhood and Adolescence (DICA) (Herjanic and Reich, 1982) to parents and selected sections (ADD, oppositional disorder [OD], conduct disorder [CD], drug use, depression, and mania) to patients.

The DICA and clinical evaluations by referring physicians were used to screen out CNS involvement, childhood autism, psychosis, physical handicaps, and uncorrected visual or auditory problems. Mental deficiency was ruled out by selecting subjects with Full Scale Wechsler IQ scores >80 , based on testing administered within 6 months of referral.

Table 1 details *DSM-III* diagnoses for which DICA criteria were satisfied in either the parent or the patient interview. It is evident that a large proportion of patients displayed OD or CD in accord with the literature. In addition, Table 2 shows that the DICA diagnoses of ADD are supported by elevated contemporaneous and retrospective childhood ratings by parents on the Home Activity Scale (Werry and Sprague, 1970) and Abbreviated Conners Hyperactivity Questionnaire (Goyette et al., 1978).

The sample was predominantly middle class (mean Hollingshead 4-point socioeconomic status = 48.80 ± 15.76 , i.e., Social Class II). Consistent with their late age of referral

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From the Departments of Psychology (Drs. Klorman, Brumaghim, and Fitzpatrick) and Pediatrics (Dr. Borgstedt), University of Rochester.

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Reprint requests to Dr. Klorman, Psychology Department, River Campus Station, University of Rochester, Rochester, NY 14627.

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TABLE 1. *Diagnostic Characteristics of the Sample*

Diagnosis	No. of Patients
Oppositional, not conduct disorder	24
Oppositional and conduct disorder	12
Not oppositional, not conduct disorder	12
Tobacco dependence	1
Alcohol use	5
Alcohol abuse	2
Marijuana abuse	1
Major depression, past	1
Adjustment disorder with depressive mood, past or present	17
Overanxious disorder, present	5
Phobia, present	5
Enuresis, past or present	16
Encopresis, past	3

TABLE 2. *Pretrial Scores on Clinical Scales*

Scales	
Home Activity Scale, retrospective	1.76 ± 0.91
Home Activity Scale, contemporaneous	1.35 ± 0.93
Abbreviated Parent Conners Questionnaire, retrospective	1.41 ± 0.70
Abbreviated Parent Conners Questionnaire, contemporaneous	1.22 ± 0.61
Parent IOWA Inattention/Overactivity Scale, contemporaneous	1.47 ± 0.71
Parent IOWA Aggression Scale, contemporaneous	1.02 ± 0.71
Parent TOTS Hyperactivity Scale, contemporaneous	-0.17 ± 0.44
Parent TOTS Attention Scale, contemporaneous	0.66 ± 0.47
Parent TOTS Aggression Scale, contemporaneous	-0.25 ± 0.55
Teacher Abbreviated Conners Questionnaire, contemporaneous	1.27 ± 0.51
Teacher IOWA Inattention/Overactivity Scale, contemporaneous	1.54 ± 0.56
Teacher IOWA Aggression Scale, contemporaneous	0.86 ± 0.58
Teacher TOTS Hyperactivity Scale, contemporaneous	-0.79 ± 0.44
Teacher TOTS Attention Scale, contemporaneous	-0.17 ± 0.40
Teacher TOTS Aggression Scale, contemporaneous	0.42 ± 0.34

Note: All values are mean ± SD. Maximum score is 4 for the Home Activity Scale and 3 for all others. Except for the Attention scale, better scores are low.

TOTS = Times on Task Scale.

and middle-class background, subjects obtained Wechsler IQs at the high end of the average range ($\bar{X} = 108.62 \pm 12.72$). However, the patients' educational achievement on the Peabody Individual Achievement Test ($\bar{X} = 98.71 \pm 12.44$) was just average for national norms and below expectation for the suburban schools that most of them attended.

Procedures

Parents and patients provided informed consent after receiving detailed verbal and printed explanations of the protocol.

Subjects received 3 consecutive weeks each of methylphenidate hydrochloride and placebo in random order and under blind conditions. Twenty-three patients were administered placebo first, and 25 were assigned to the opposite order. Both substances were dispensed in capsules of identical appearance. Placebo capsules contained lactose and small amounts of artificial yellow dyes and quinine, in order to approximate the color and taste of commercially available methylphenidate.

The dose of methylphenidate was based on body weight. Subjects weighing under 37.5 kg received 7.5 mg in the morning and at noon in the first week, 10 mg b.i.d. (morning and noon) in the second week, and 10 mg b.i.d. plus 5 mg at 4 P.M. in the third week. Subjects weighing between 37.5 and 54.5 kg received 10 mg b.i.d. in Week 1, 12.5 mg b.i.d. in Week 2, and 12.5 mg b.i.d. plus 7.5 mg at 4 P.M. in Week 3. Subjects weighing above 54.5 kg received 12.5 mg b.i.d. in Week 1, 15 mg b.i.d. in Week 2, and 15 mg b.i.d. plus 10 mg at 4 P.M. in Week 3.

The morning and late afternoon doses were administered at home and the noon dose at school. Parents were telephoned weekly to foster compliance and to check on possible side effects. The school nurse was also contacted at the start of the protocol in order to promote cooperation and to encourage prompt reports of noncompliance if it occurred. In fact, patients missed only occasional doses, and counts of unused capsules at the end of each phase indicated that compliance was good.

Reports of significant side effects led to blind lowering of dosages for four patients (three females) in the active phase and two (two males) in the placebo phase. The mean dose for the third week of methylphenidate treatment was 35.21 ± 5.94 mg/day. The sample's mean weight was 54.91 ± 13.94 kg, so that the terminal dose was 0.64 mg/kg/day.

Clinical Outcome Measures

Parents and teachers completed the Abbreviated Conners Hyperactivity Questionnaire plus the Iowa Inattention/Overactivity and Aggression scales (Loney, 1987) on a weekly basis and the Hyperactivity, Attention, and Aggression scales of the Time on Task Scale (TOTS) (Loney, 1987) at the end of each phase. Finally, teachers and parents were requested to respond in writing each week to open-ended questions about the patient's behavior. Parents were asked about involvement in schoolwork, general deportment, ability to concentrate, and relations with others. Teachers were queried about academic work, behavior in class, and general attitude toward school. In addition, both parents and teachers were given space to write about any other matter they wished. If a patient had more than one teacher, the authors identified two or, if available, three teachers who were familiar with him and taught a class in which he had academic or behavior problems. These teachers were telephoned at the start of the protocol in order to stress the importance of completing questionnaires throughout the trial.

The patient reported for three, individual, 3-hour appointments held at noon shortly before the trial as well as on the last day of each phase. In the first two meetings, parents were given dual sets of blank rating scales and vials containing

TABLE 3. Percentage of Patients with Side Effects

Symptoms	Parent Report		Patient Report	
	Placebo	Methylphenidate	Placebo	Methylphenidate
Appetite loss	10.4	27.1**	6.4	29.8***
Increased thirst	4.3	2.1	8.7	13.0
Dry mouth	2.1	8.3	8.5	19.1*
Stomachaches	6.2	6.2	4.3	8.5
Nausea	2.1	0.0	2.1	2.1
Headaches	6.4	12.8	4.3	6.4
Sleep problem	14.6	16.7	12.8	17.0
Shakiness	0.0	0.0	2.1	8.5*
Crying	8.5	4.3	0.0	6.4
Anger	34.8	23.9	17.4	10.9
Unhappiness	22.9	20.8	12.8	10.6
Sadness	4.2	4.2	0.0	4.3

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$ compared with placebo.

capsules for the ensuing phase for home and school, respectively. In the last two visits, the patient and parent were questioned separately by a psychologist (JTB or RK) about the occurrence of side effects listed in the Subject's Treatment Emergent Symptom Scale (Guy, 1976) during the preceding treatment phase. The interviewer coded whether each symptom was present to a marked degree or absent. For statistical analyses, the authors counted the occurrence, separately for each phase and for patients and parents, of 12 symptoms identified in Table 3 and selected a priori as sensitive to stimulant therapy. Finally, the patient was weighed without shoes and outer garments on a professional scale by a staff member, blind to drug condition.

Next, the patient received his noon dose and participated in a cognitive evaluation similar to Coons et al.'s (1987). (These results will be reported in a separate publication.) The patient was asked to complete the Nowlis Mood Scale (Nowlis and Green, 1965) immediately before his dose as well as 1.25 and 3 hours later (or the equivalent points in the pretrial session). In a practice session before the trial as well as in all subsequent sessions, the patient was offered definitions in simple English of the items of the Nowlis Mood Scale. Statistical analyses were focused on five factors previously (e.g., Strauss et al., 1984) found to be sensitive to the effects of stimulants: vigor, elation, surgency, fatigue, and concentration.

Another measure obtained during this session consisted of ratings of the patient's behavior by his examiner on eight selected items (withdrawn/unspontaneous, negative/resistant/uncooperative, crying, fidgety/hyperactive, distractible, silly/inappropriate, loud voice, low voice) adapted from the Child Psychiatric Scale (Guy, 1976). Specific behavioral criteria (available from the authors) were provided to the examiner, and the ratings (0 = not at all; 3 = extremely) were averaged for each session.

In the last session, the parent and patient were asked separately by a psychologist (JTB or RK) to (a) state which phase of treatment yielded better outcome, and (b) rate, using the Loney and Ordon (1975) scale (Table 4), the degree of global improvement observed in this phase in comparison to the period before the trial. At this point, the blind was broken.

TABLE 4. Parents' and Patients' Ratings of Global Outcome

Rating	Score	% of Parents	% of Patients
Dramatic response	7	0.0	4.4
Total improvement	6	4.2	2.2
Good response	5	22.9	19.6
General improvement	4	22.9	10.9
Some consistent improvement	3	2.1	0.0
Mixed or variable response	2	10.4	13.0
Unimpressive response	1	4.2	4.4
No change at all	0	10.4	30.4
Dramatically bad response	-3	22.9	15.2

Note: Loney-Ordon ratings by parents and patients for the phase that each found better. If the placebo was chosen, a rating of dramatically bad response was entered.

Scoring

Because no scoring keys are available for the TOTS parent scales, the authors used those for the teacher scales, as suggested by the developer of the scales (Loney, personal communication, 1988).

The weekly comments of teachers and parents were scored by an author (PAF), blind to the patient's identity and drug order. Comments were scored separately for the two treatment phases and parents/teachers, but all data for a particular respondent were scored at the same time. The responses to an entire question were scored +1 if they were primarily positive; 0 if they were blank, ambiguous, neutral, or irrelevant; and -1 if they were negative. For each week of treatment, the authors obtained the algebraic mean of these ratings or valence. Another author (JTB) scored 12% of these comments and agreed with the first judge on 87% of the items rated, excluding unanswered questions so as to not inflate this estimate of reliability.

Teacher ratings and comments were averaged over all participating teachers who provided responses for at least 2 of the 3 weeks for both phases of the trial. Any missing data were estimated by the average of that teacher's two other

ratings for the same phase. To establish the degree of concordance among teachers, the authors computed intraclass correlations (Shrout and Fleiss, 1979) for each of the 6 weeks of the protocol (or the last week of each phase for TOTS scales) separately for patients with three teachers and those with at least two teachers. The reliabilities for rating scales were acceptable, although those for Aggression were somewhat lower. The correlations based on two teachers ranged from: 0.31 ($p < 0.04$) to 0.50 ($p < 0.001$) for the Abbreviated Conners Questionnaire; 0.33 ($p < 0.03$) to 0.42 ($p < 0.006$) for the IOWA Inattention/Overactivity Scale; 0.08 (NS) to 0.47 ($p < 0.002$) for the IOWA Aggression Scale; 0.50 ($p < 0.001$) to 0.52 ($p < 0.004$) for the TOTS Hyperactivity Scale; 0.33 ($p < 0.03$) to 0.40 ($p < 0.009$) for the TOTS Attention Scale, and 0.14 (NS) to 0.27 ($p < 0.06$) for the TOTS Aggression scale. The valence of teachers' comments had considerable lower reliability than their ratings, ranging from -0.16 (NS) to 0.35 ($p < 0.05$). However, reliability estimates for these open-ended measures are affected by (a) inconsistent idiosyncratic response tendencies among teachers, e.g., propensity to leave questions blank or insert irrelevant comments; and (b) highly individualized events or impressions which may differ from class to class.

Results

The data were subjected to univariate and multivariate analyses of variance with age (above/below the median of 13.7 years) and drug order as group factors and repeated measures on drug treatment and other factors, as appropriate (e.g., weeks of treatment, time in the session, and respondent). Covariance adjustments were performed based on pretrial scores for rating scales and on precapsule ratings for the Nowlis Mood Scale. There were very few significant interac-

tions of drug treatment with drug order or age, and the occasional exceptions are listed below.

Somatic Effects of Treatment

Figure 1 shows that subjects weighed 0.83 kg less under methylphenidate than placebo, $F(1,39) = 20.37$, $p < 0.001$. In comparison with pretrial values, weight decreased under methylphenidate ($p < 0.05$ by Tukey's HSD test) and non-significantly increased under placebo.

As shown in Figure 1 and Table 3, there was a slight elevation in side effects for methylphenidate versus placebo, but mostly for patients' reports, drug $\times$ parent/patient $F(1,43) = 4.00$, $p = 0.06$. In fact, the difference between methylphenidate and placebo was significant for patients ($p < 0.01$ by Tukey's HSD test), but not for parents. As shown by McNemar's (1962) chi-square test of change, patients made more complaints during stimulant than placebo treatment for appetite loss, $\chi^2(1) = 8.07$, $p < 0.005$, dry mouth, $\chi^2(1) = 3.57$, $p < 0.06$; and shakiness, $\chi^2(1) = 3.00$, $p < 0.10$. Parents also reported appetite loss more often for methylphenidate $\chi^2(1) = 4.00$, $p < 0.05$.

Weight loss during methylphenidate treatment scored, relative to the immediately preceding measurement period, was not significantly correlated with reports of appetite loss by parents, $r(41) = 0.05$; or patients, $r(41) = 0.01$.

Subjective Mood

Figure 2 displays mood ratings over the course of each session. A multivariate analysis showed significant differences among treatment, $F(5,35) = 3.57$, $p < 0.02$. Separate

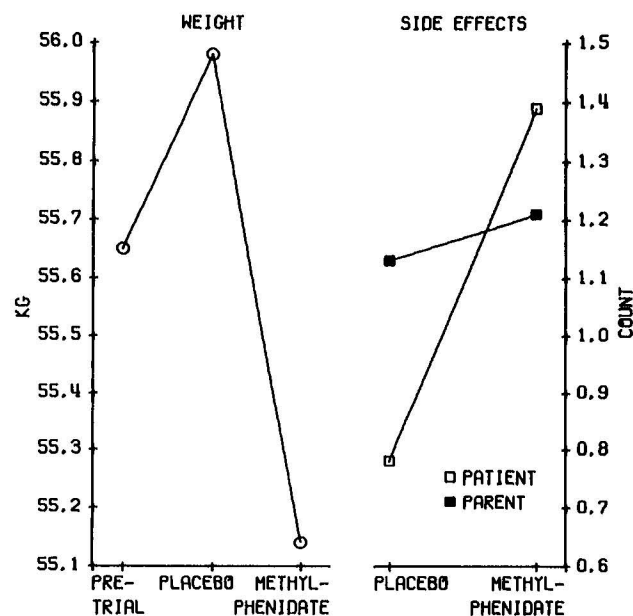


FIG. 1. Left: Weight pretrial and after 3 weeks of placebo and 3 weeks of methylphenidate. Right: Count of selected side effects (See Table 3) after 3 weeks of placebo and 3 weeks of methylphenidate.

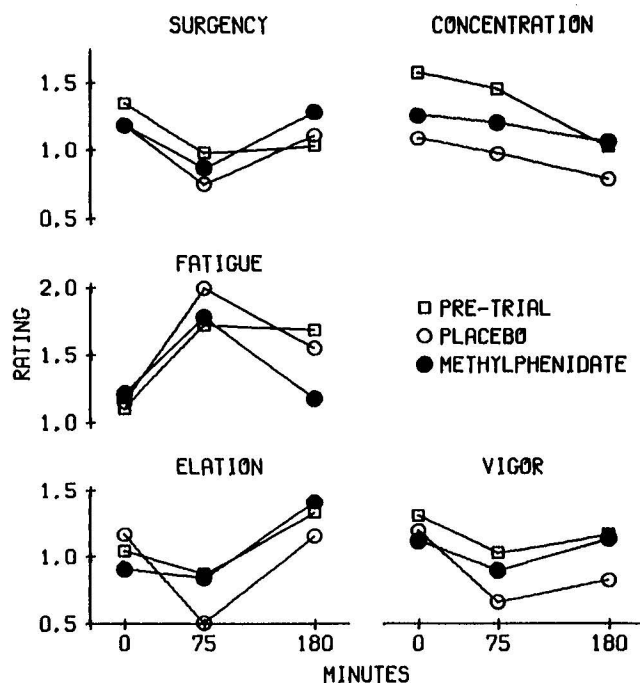


FIG. 2. Nowlis mood ratings (0 = Not at All; 4 = Extremely) by ADD adolescents at the start of each session (before ingestion of a capsule), 75 and 180 minutes later. Data are adjusted for covariance on precapsule ratings.

analyses of each mood factor confirmed that mood was higher under methylphenidate than placebo for surgency, $F(1,38) = 2.89, p < 0.10$; concentration, $F(1,38) = 6.43, p < 0.02$; fatigue, $F(1,38) = 5.92, p < 0.02$; elation, $F(1,38) = 7.40, p < 0.01$; and vigor, $F(1,38) = 8.21, p < 0.007$.

The elevation of mood by the stimulant generally did not differ for older versus younger patients, except that elation ratings were enhanced by methylphenidate only for younger patients, drug $\times$ age $F(1,38) = 4.53, p < 0.04$.

Irrespective of treatment, over the course of the session, there was an increase in surgency, elation, and vigor and a decrease in fatigue and concentration ($p < 0.02$ or less). These findings may reflect a pharmacologically nonspecific sense of relief and decreased mental energy after completing a lengthy and taxing test session.

Clinical Outcome Measures

Figure 3 displays weekly ratings by teachers and parents.

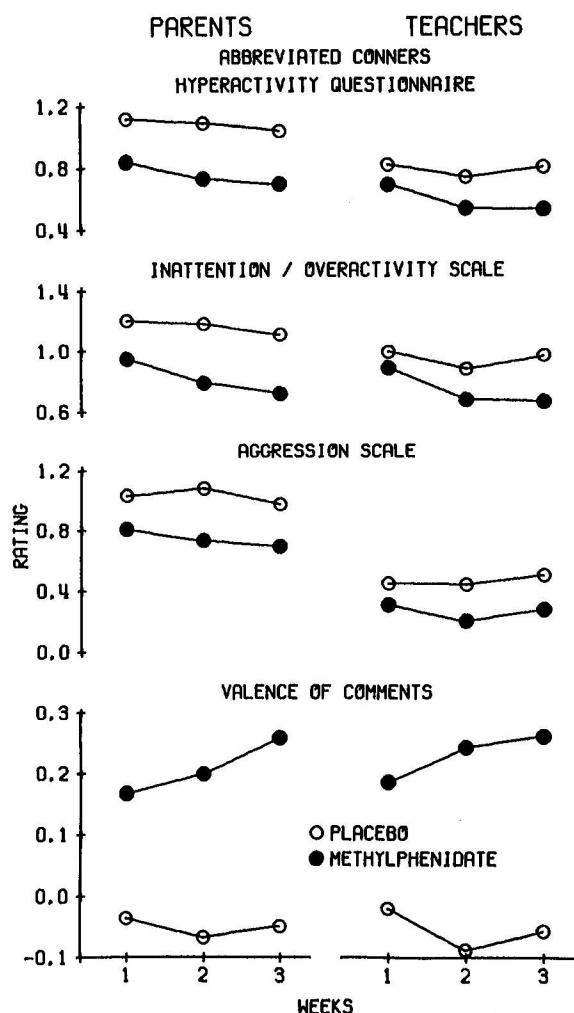


FIG. 3. Ratings by parents (left) and teachers (right) for each week of placebo and methylphenidate treatment. Higher scores indicate worse adjustment for the Abbreviated Conners Questionnaire, Inattention/Overactivity Scale, and Aggression Scale and better adjustment for valence of comments. Scores on scales are adjusted for covariance on pretrial ratings.

A multivariate test indicated that patients' behavior was rated as significantly better under methylphenidate than under placebo, $F(4,38) = 5.78, p < 0.001$. Although parents rated patients as more disturbed overall than did teachers, multivariate $F(4,38) = 6.37, p < 0.0005$, both informants described a comparable degree of benefit for drug treatment, multivariate parent/teacher $\times$ drug $F(4,38) = 0.60, p < 0.67$. Separate analyses of each scale showed that parents reported significant improvement under stimulant treatment for the Abbreviated Conners Questionnaire, $F(1,44) = 13.90, p < 0.0005$; Inattention/Overactivity scale, $F(1,44) = 12.28, p < 0.002$; Aggression scale, $F(1,44) = 8.57, p < 0.006$; and valence of comments, $F(1,40) = 8.19, p < 0.007$. For the Abbreviated Conners Questionnaire and Inattention/Overactivity scales, parents detected significantly less disturbance over weeks ($p < 0.003$ for Weeks' linear trend) for the two treatments combined. These effects reflect changes in parents' perceptions due to repeated ratings but independent of treatments.

Similar results were obtained for teacher ratings, which indicated significant improvement for the Abbreviated Conners Questionnaire, $F(1,39) = 14.48, p < 0.0005$; Inattention/Overactivity scale, $F(1,39) = 9.18, p < 0.005$;

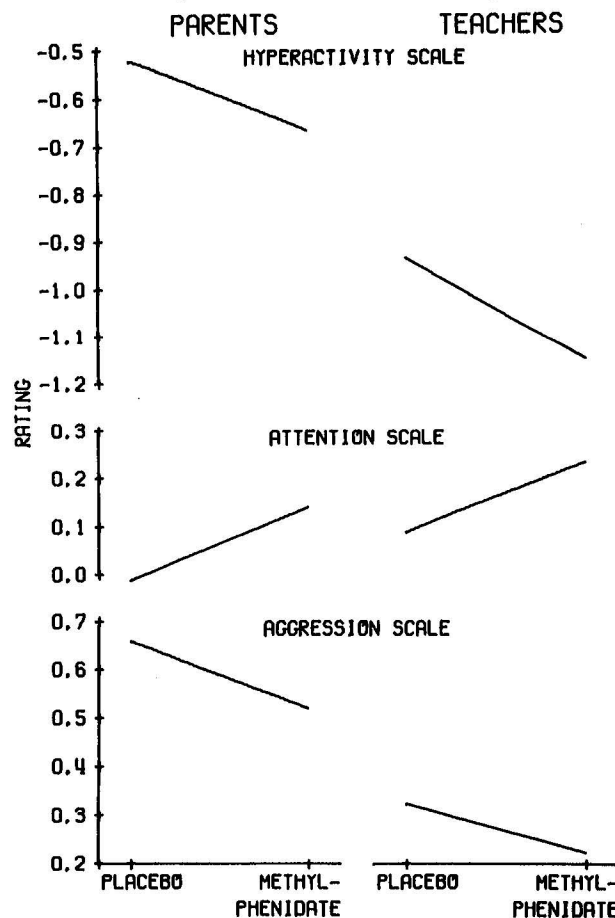


FIG. 4. Ratings by parents (left) and teachers (right) after 3 weeks of placebo treatment and 3 weeks of methylphenidate therapy. Higher scores indicate worse adjustment for the Hyperactivity and Aggression Scales and better behavior for the Attention Scale. Data are adjusted for covariance on pretrial ratings.

Aggression scale, $F(1,39) = 11.52, p < 0.002$; and valence of comments, $F(1,37) = 20.26, p < 0.0001$. Teachers reported greater improvement as dosage increased over the course of the methylphenidate phase (drug $\times$ weeks linear trend interaction) for the Abbreviated Conners Questionnaire ($p < 0.03$), Inattention/Overactivity scale ($p < 0.007$), and valence of comments ($p < 0.06$). For aggression ratings, teachers reported greater improvement for younger than older patients, drug $\times$ age $F(1,39) = 5.46, p < 0.03$.

Figure 4 displays the results for the TOTS scales, which were administered only at the end of each phase. As found for weekly ratings, parents' TOTS ratings revealed greater disturbance overall than did those of teachers, multivariate $F(3,38) = 12.11, p < 0.0001$. However, both informants reported improvement under methylphenidate, multivariate $F(3,38) = 3.85, p < 0.02$, to a comparable degree, multivariate parent/teacher $\times$ drug $F(3,38) = 0.76, p = 0.40$. Separate analyses of parents' ratings disclosed improvements in aggression, $F(1,44) = 5.30, p < 0.03$; hyperactivity, $F(1,44) = 3.96, p = 0.05$; and attention, $F(1,44) = 3.71, p = 0.06$. Similarly, teachers reported reductions of aggression, $F(1,39) = 5.67, p < 0.03$; hyperactivity, $F(1,39) = 17.16, p < 0.0002$; as well as increased attention, $F(1,39) = 4.97, p < 0.04$.

As shown in Table 4, ratings of global outcome by parents and patients were similar, $F(1,44) < 1, p = 0.69$. In combination, these judgments indicated improved outcome, as they differed significantly from the expected value of zero, $F(1,44) = 27.84, p < 0.0001$. Unexpectedly, better ratings were found for patients who received methylphenidate first ($\bar{X} = 2.70$) than for those administered placebo first ($\bar{X} = 1.08$) $F(1,44) = 5.29, p < 0.03$. It is tempting to interpret this solitary order effect as a chance finding. In any event, it should be noted that a positive outcome was reported for patients treated in both orders, although with differing levels of enthusiasm.

The examiner's mean ratings for the three tests sessions were 0.10 ± 0.15 for pretrial, 0.12 ± 0.16 for placebo, and 0.06 ± 0.06 for methylphenidate. The analysis of variance disclosed that ratings for methylphenidate were significantly superior to those for placebo, $F(1,44) = 8.66, p < 0.006$.

Subsidiary Analyses by Comorbidity

Comparisons of patients with secondary diagnoses of oppositional and conduct disorder versus neither indicated that the clinical benefits of methylphenidate therapy were comparable among these three diagnostic subgroups. This similarity is reflected statistically in nonsignificant interactions of drug $\times$ subgroups for weekly clinical measures, multivariate $F(4,37) = 0.58, p < 0.68$; TOTS ratings, multivariate $F(3,38) = 0.96, p < 0.43$; and ratings of global outcome, $F(2,45) = 0.43, p < 0.65$. Similarly, comparisons of patients with and without secondary diagnoses of major depression/adjustment disorder with depressive mood disclosed comparable gains of treatment for weekly scales, multivariate $F(4,38) = 0.85, p = 0.77$; TOTS ratings, multivariate $F(3,39) = 0.47, p = 0.70$; and global outcome ratings, $F(1,46) = 0.00, p = 0.99$.

These results suggest that for the present sample, the benefits of methylphenidate were comparable across the conduct/oppositional and depressive spectra. However, a more extensive sampling of these secondary diagnoses is needed to definitively rule out their interaction with treatment effects. There were relatively few ADD patients diagnosed CD or not oppositional/not conduct disorder. Similarly, only one of the patients was diagnosed major depression, and the diagnoses of adjustment disorder with depressed mood should be confirmed by a later interview in order to establish a more secure diagnosis. Thus, it cannot be ruled out that a larger sample of these subgroups may reveal differential response to stimulant treatment.

Discussion

Somatic Effects

The weight loss resulting from methylphenidate treatment is consistent with follow-ups of prepubertal ADD children for 1 or more years (e.g., Satterfield et al., 1979; Mattes and Gittelman, 1983). In accord with these reports, the authors did not find weight loss to be correlated with appetite loss. Although most research has shown that stimulant induced suppression of growth and weight is partly or fully overridden by discontinuation of therapy (Safer and Allen, 1973; Mattes and Gittelman, 1983), there is a need to study stimulant effects on stature and weight during puberty and the concomitant growth spurt. Thus, it would be important to perform longer follow-ups of stimulant-treated adolescents that include weight as well as height measurements.

Relatively few side effects were reported, in part, because of the adjustment of dosages in response to these symptoms. In fact, parents and patients were directed to rate side effects present at the end of treatment, i.e., after any dosage reductions or adaptation to medication. It is conceivable that this procedure attenuated somewhat reports of side effects. On the other hand, the authors' approach was intended to focus on the terminal dosage, which best represented the present treatment.

The lower rate of side effects reported by parents rather than patients may be due to the facts that many of these symptoms were not directly observable by parents, and adolescents may not disclose them to their parents. In accord with this hypothesis, the only significant side effect reported by parents was lower appetite, a symptom which is more noticeable than dry mouth or occasional tremors.

Subjective Mood

There is very limited evidence on the impact of stimulants on ADD children's mood (Cantwell and Carlson, 1978). Although therapists and parents have described euphorogenic effects (Conners et al., 1965), children's self-ratings have not confirmed these findings (Rapoport et al., 1980), perhaps because of the difficulty in validly assessing children's moods. In a sample of bright, normal children, methylphenidate significantly elevated self-reported mood in comparison to placebo during tests of cognitive functioning (Peloquin and Klorman, 1986). Thus, it is likely that a similar effect might be found in ADD children if assessment difficulties were

overcome. In fact, the mood elevation by stimulants in the present as well as previously studied adolescent ADD patients (Klorman et al., 1987) parallels findings for normal adults and normal children. Presumably, the reason that mood elevation has not been reliably obtained in stimulant-treated ADD children is that they are less adept than adolescents at describing their mood.

The elevation of patients' mood by methylphenidate was of moderate extent, roughly a quarter of a scale unit. However, mood effects in normal adults and children also have been of limited magnitude (e.g., Strauss et al., 1984). Thus, both the direction and degree of mood change suggest that ADD adolescents have a nonparadoxical, normal affective response to stimulants.

Although mood was definitely improved by methylphenidate in comparison to placebo, the stimulant did not elevate mood in comparison with the drug-free pretrial assessment. This result should be viewed in the context of a relatively taxing test session, such that subjects become less cooperative and cheerful in later sessions (Klorman et al., 1987, 1988). Thus, methylphenidate restored mood to levels obtained during the initial, drug-free assessment.

Because mood was evaluated only in the laboratory, it cannot be concluded that comparable mood elevation was necessarily present at home or school. However, the results are consistent with that possibility, insofar as previous studies and clinical reports describe improved mood in naturalistic settings.

Clinical Outcome

There was consistent and widespread evidence of improved behavior under methylphenidate. Parents, teachers, and the examiner rated the patients as significantly improved under methylphenidate. These gains were found in ratings as well as open-ended comments such as those which might be elicited in a clinical interview. To the extent that halo effects did not operate, the amelioration of behavior was described in terms of reductions in activity level, inattention, as well as aggression/oppositionality. The same wide range of behavioral domains improved by stimulant treatment has been found in work with prepubertal ADD patients (Gittelman and Kanner, 1986).

The present results differed from a previous report of more pronounced benefits of methylphenidate in reports by parents than teachers of ADD adolescents (Klorman et al., 1987). In fact, for the present sample, teachers and parents were equally sensitive to the effects of medication. Moreover, only teachers reported additional improvement as the dosage of methylphenidate was raised over the 3 weeks of treatment. This difference may reflect greater sensitivity by teachers because of their access to behaviors directly affected by stimulant medication as well as their interaction with the patient during the time that the drug is most active. Teachers' attentiveness to the patients may have been maximized by their conduct and oppositional disorders, which, in combination with academic difficulties, often prompted a referral from the school.

Nevertheless, teachers consistently rated the patients considerably lower than did parents on standard scales of hyperactivity and associated disorders. In fact, despite the patient

clinical disturbance of these ADD adolescents, they often did not stand out to teachers as having attentional difficulties. Rather, teachers often viewed the patients as having "motivational" or "emotional" problems. This discrepancy between teachers and parents might suggest that these ADD patients had a milder form of the disorder, a fact that could explain their lack of prior stimulant therapy. However, stimulant therapy had, in fact, been recommended for several patients by professionals and teachers in earlier grades. In other cases, parents and past teachers had not been impressed with the patients' academic underachievement, because they avoided outright failures, perhaps by virtue of their relatively high IQ.

The effectiveness of stimulant therapy for ADD adolescents offers a treatment alternative to the enduring disturbances of these patients after discontinuation of this treatment. Every indication is that the gains of stimulant therapy are not preserved after treatment ends. Thus, the continuing efficacy of these medications in adolescence supports maintenance treatment as long as there is evidence of clinical effectiveness. At the very least, stimulant therapy should not be terminated routinely on the assumption that it is not effective for this age group.

Yet, despite the benefits of stimulant therapy, the magnitude of the results was more limited than those found in younger patients. For instance, nearly 40% of parents described the outcome of treatment as unimpressive or worse. In addition, the improvement in clinical ratings, although statistically significant, is of moderate degree. It may be argued that these results are due to inherently lower effectiveness of stimulants for adolescents, perhaps because of biological factors or the greater gravity of associated problems with aggression and motivation in this age range. It seems implausible that adolescence produces reduced responsiveness to stimulants, because of evidence that ADD adults respond to this treatment (Wender et al., 1985). Similarly, it is difficult to invoke entrenched secondary disorders as an explanation for moderate effectiveness of stimulants, in view of the comparable results for patients with and without oppositional or conduct problems.

Another explanation which might be offered is that the present dose schedule was inadequate. However, the authors found significant weight loss and side effects that for some patients required lowering of dosages. Thus, the likelihood is that higher dosages would not have been tolerated by most patients.

An alternative interpretation of the results is that stimulants are equally effective for prepubescent and adolescent ADD patients, but that parents and teachers, on whom we depend for detecting clinical outcome, have greater difficulty in discerning these effects. Adolescents spend much less time at home or in any single classroom than do children. As a result, teachers and parents may be less well attuned to adolescents' behavior and, therefore, less sensitive to treatment effects than for younger patients. In fact, parents often expressed uncertainty about describing patients' behavior during the trial because of their limited interaction. Moreover, the present protocol provided afternoon doses only in the last week of treatment, so that there was a relatively brief period in

which parents could observe patients before the drug had metabolized. This reasoning generates the hypothesis that a longer follow-up is required in order for parents and teachers to detect improvements resulting from stimulant treatment in ADD adolescents. The authors had several informal reports by parents of patients who continued in stimulant therapy to the effect that the gains of treatment were more pronounced after a few months. Once again, these observations argue for a longer follow-up in order to more clearly delineate the range of treatment effectiveness.

Although the clinical effects of methylphenidate treatment were of moderate degree, it should not be overlooked that there was significant behavioral improvement in a population that has serious conduct and cognitive problems in adolescents. The present results suggest that these difficulties can be attenuated by stimulant therapy.

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